

Frequency-Domain Denoising

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Implementing and comparing Fourier-domain restoration methods against spatial-domain baselines for denoising

Image source

Clean reference images are taken from **CBSD68** (68 colour natural images, 321×481), obtained from the benchmark collection at https://github.com/majedelhelo/denoising_datasets. CBSD68 is the standard colour image-denoising benchmark, derived from the Berkeley Segmentation Dataset (BSDS500), so the PSNR and SSIM values reported here are directly comparable with published restoration results. The collecting repository states that the datasets are gathered for convenience, with no ownership claimed.

The same 68 base images are used for the deblurring experiments, so that differences in the reported metrics reflect the degradation and the restoration method rather than the image content.